Food Access in North Carolina

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Introduction

Background & Relevant Work

There are 18.8 million Americans living in food deserts today — areas with limited access to affordable and nutritious food. The USDA has defined food deserts to be census tracts that have a poverty rate of 20% or greater (or median family income that is less than 80% of statewide family income) and where at least 33% of the population live more than 1 mile from the supermarket in an urban area/ 10 miles in a rural area. Food deserts are created by distance from healthy food sources, usually influenced by structural decisions that have historically limited healthy food access for minority communities. Understanding this issue requires examining how redlining policies institutionalized neighborhood disinvestment, disproportionately exposing low-income and minority communities to less healthy food environments. These disparities carry serious consequences: food deserts are linked to worse health outcomes, including shorter life expectancy and a higher risk for adverse cardiovascular events. Given these health risks and the disproportionate racial impact of food deserts, it is critical to evaluate what factors contribute to their persistence and how current policy can work toward reversing years of racial residential segregation.

This issue is particularly relevant to us as Duke students, as 9 of the top 10 states with the greatest share of residents living in food deserts are in the South. In particular, North Carolina is one of the top 10 agricultural states in the US, but its food insecurity rate is 13.9% — 2.2 percentage points higher than the national average. This proximity makes understanding the structural drivers of food deserts a matter of community concern. Thus, I seek to answer the following question: what features can best explain whether a census tract is a food desert (classified as a low-income, low-access area) in North Carolina?

Studies have found that neighborhood racial composition and poverty rates are associated with food store availability. Interestingly, this study notes that these associations vary by rural and urban regions, which is why interventions should be developed locally rather than universally. A cross-sectional survey of randomly selected households in public housing communities in

Baltimore, MD found that individuals with a personal vehicle were more likely to report ease in reaching health food stores as compared to others.

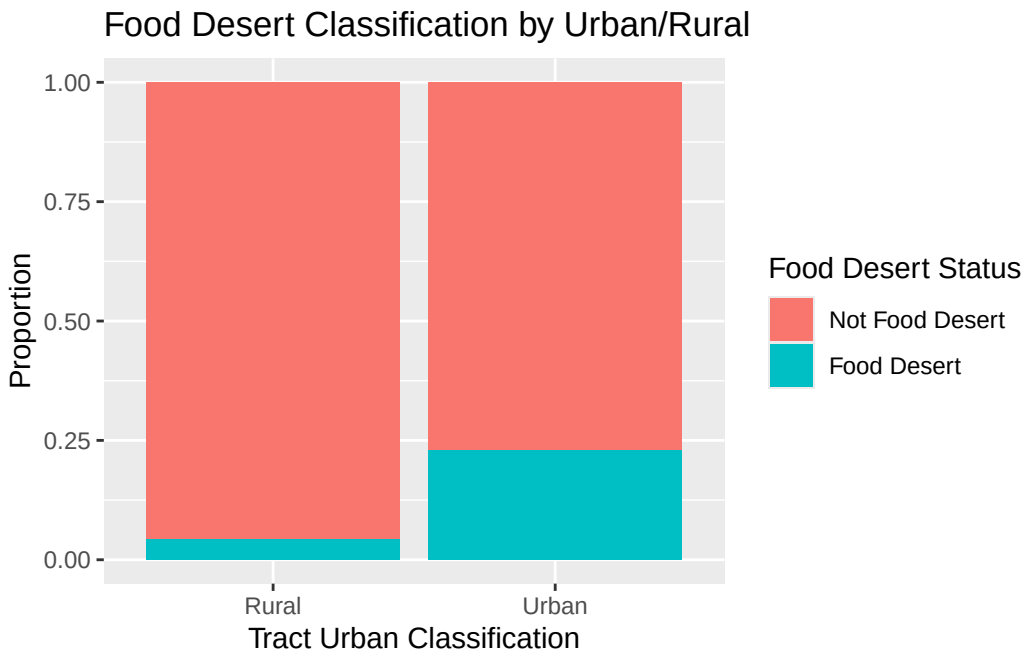
Data & Key Variables

I will use the USDA's 2019 Food Research Atlas dataset to investigate my question and choose key predictors to understand patterns of food access in North Carolina. The 2019 Food Access Research Atlas provides an overview of food access indicators in the US by measuring variables like supermarket accessibility, race, poverty rate, and median income. This dataset is a combination of data collected by the 2010 Decennial Census, the 2014-18 American Community Survey (ACS), and two 2019 lists of supermarkets, one being from Trade Dimensions TDLinx. Observations are at the census tract level and therefore may not be independent, as nearby census tracts are likely to exhibit similar patterns in food desert status and food desert predictors. To address this limitation, I will take a random sample of 20% from the North Carolina subset of the dataset. Appendix A confirms that this random sample truly represents the distribution of food deserts across the state and therefore allows our subsequent analysis to be generalizable to the entire state of NC.

There is also substantial missing data in the `LALOWI1_10` variable, which represents the amount of people living in a low access, low income area. Thus, I will use the `LILATracts_1AND10` response variable, which is an indicator variable of what `LALOWI1_10` measures — whether a tract is classified as low income and low access (food desert). The `MedianFamilyIncome` variable also has 17 missing observations. However, since this is not a significant portion of the data for the NC sample (17 missing out of 448 observations), I will delete these observations rather than impute counts, so as not to assume patterns for certain areas. As a result, I also acknowledge that this data may not be entirely representative of the entire state, especially if those null values were concentrated in a specific area.

To understand factors that contribute to a census tract's categorization as low-income and low-access (or being a food desert), I will focus on the `LILATracts_1And10` response variable as a proxy measure for food desert (as it fits the USDA's definition of a food desert) and the following predictor variables: `Urban` , `TractSNAP` , `TractHUNV` , `PovertyRate` , `MedianFamilyIncome` . `Urban` is an indicator variable that classifies a census tract as urban or not using the official definition by the US Census (urban area has more than 2,500 people). This predictor will contribute to understanding whether food deserts vary by urbanicity, which is helpful in understanding where policymakers should direct resources to address such issues. Next, `TractSNAP` will detail the number of housing units within a census tract receiving SNAP benefits. The rationale behind including this predictor is that communities with more individuals receiving SNAP benefits are also likely to be under resourced, characteristic of food deserts. `TractHUNV` provides information on number of housing units with access to a vehicle, which could illuminate the low access to grocery stores aspect of the food desert definition. Finally, `MedianFamilyIncome` and `Urban` will both be used to determine the general economic state of families in a census tract.

Exploratory Data Analysis

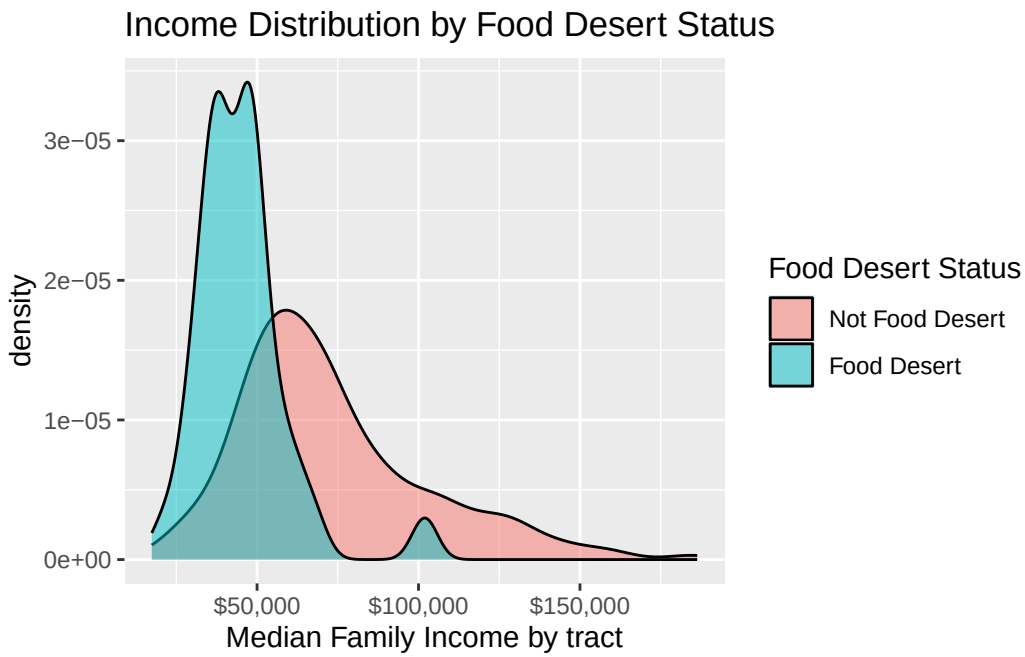


The map illustrates how the majority of North Carolina does not contain areas classified as food deserts. More specifically, it shows that food deserts are concentrated in small areas around the state, and viewed in conjunction with the stacked bar chart, we understand that census tracts classified as food deserts are primarily also categorized as urban areas. To be specific, in the representative random sample of census tracts in NC, about 84% are not food deserts. A similar pattern is evident among all census tracts in NC, where about 87% of tracts are not food deserts. Due to the class imbalance in this variable, it is likely that the model fit on this data will not be accurate at predicting tracts that *are* food deserts, so I will gear my model selection toward explaining factors that contribute to food deserts.

Each of the quantitative predictors (`PovertyRate`, `TractSNAP`, `TractHUNV`, `MedianFamilyIncome`) have a unimodal and slightly right-skewed distribution. `Urban` shows a similar pattern of positive skew as the majority of census tracts in North Carolina (around 62%) are classified as urban. To ensure that the relationships between the quantitative predictors and the response is linear, I plotted each against the predicted probability of being a food desert and confirmed linear relationships for each of the predictors except poverty rate in Appendix B. This helped me determine to log-transform the poverty rate when fitting my logistic regression model.

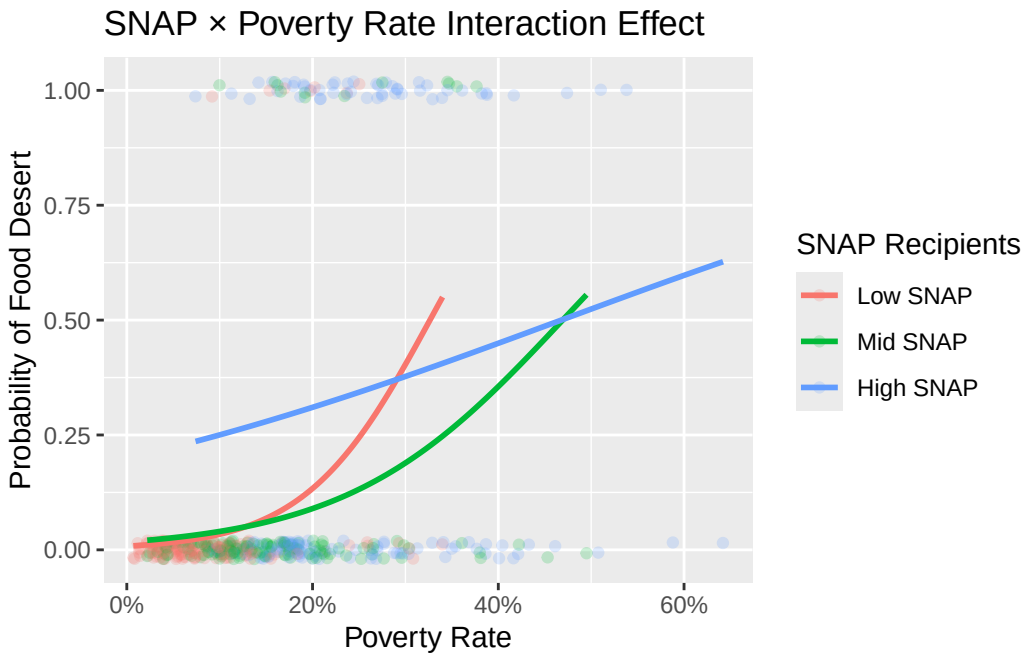
Due to this skew, the median is a good indicator of the center of the distribution for each of the variables. Thus, we see that the median poverty rate in NC is about 14.30%, which is higher than the United States' official 10.6% poverty rate. There are also some insignificant outlying census tracts with poverty rates above 50%. On average, 188 housing units in a census

tract receive SNAP benefits. The IQR for this distribution is 255 housing units, which seems relatively large, but considering the fact that a census tract can range anywhere from 1,200 to 8,000 people, this spread is understandable. Traditionally, 80 housing units in a census tract do not have access to a vehicle. This is surprisingly low but realistic since a car is a necessity in most parts of NC due to the state's lack of public transportation. The relatively smaller spread (IQR=11) is in line with our understanding of NC's public infrastructure and the necessity of a vehicle, regardless of income status. Finally, `MedianFamilyIncome` has a distribution very similar to `PovertyRate` , which is not unexpected since both are roughly measuring the economic state of a census tract. There is a center around $61,500 median family income per census tract, much lower than the US's $83,000 median household income. The spread of the distribution (IQR) is about $20,000, with outliers above median family incomes of $150,000.



The double-density plot reveals that median family income is more variable among food desert tracts than non-food desert tracts. For instance, tracts that are not food deserts have a similar center to tracts that are food deserts, but the median incomes are more spread out from $45,000 to $125,000, while median incomes for food desert tracts are mostly concentrated around $45,000 to $50,000. This distinction suggests a strong association between median family income and food desert status.

Interestingly, when observing the relationship of the quantitative predictors and the response variable, there are more outliers and a higher spread for non-food desert tracts. This may be due to less uniformity in socioeconomic characteristics among non–food desert tracts.



My key variables assess the geographic makeup of a tract (`Urban`) as well as poverty or economic status of individuals within the tract (`MedianFamilyIncome` , `PovertyRate` , `TractHUNV` , and `TractSNAP`). Since SNAP benefits are determined by relative income level and poverty rate, I visualize the interaction between the number of households receiving snap benefits in a tract and the tract poverty rate. According to the plot, there appears to be an interaction effect, as the slopes of the lines — representing different levels of snap receipt — differ. Most interesting is that tracts with a high number of SNAP recipients do not see as drastic an increase in the probability of being a food desert as poverty rate increases, as compared to tracts with a low number of SNAP recipients. This pattern could be explained by the fact that if a tract has a high number of households receiving SNAP benefits, then it is already more likely to be classified as a food desert, hence the flatter slope as poverty rate increases. For tracts with a low number of households receiving SNAP benefits, poverty rate then serves as an important metric of food desert status, which could explain the sharp increase in probability of being a food desert with each 1 percentage point increase in poverty rate.

In working with health and social science data, I am also cognizant of confounding being a limitation. This dataset does not provide a comprehensive account of all the factors that impact whether a census tract is a food desert since other variables can influence this measure/ the measure of other predictors. For instance, the `TractHUNV` predictor measures the number of housing units without a vehicle in a certain tract. Though the absence of a vehicle can make healthy food inaccessible, and therefore contribute to the categorization of an area as a food desert, the variable does not give a complete picture of accessibility — especially since access to public transportation is not considered. I will remain mindful of these limitations and incorporate them into my interpretation of the model to ensure that my conclusions are

statistically sound and representative of the data.

Methodology

We will fit a logistic regression model to predict what factors contribute to food deserts across counties in North Carolina. This model is best suited for my dataset because I wish to predict whether a specific census tract is or is not classified as a food desert. Since this is represented as a probability of being a food desert (binary choice), logistic regression will help make such a prediction. Before fitting the model, I will log-transform the poverty rate in order to satisfy the linearity condition between the response and predictor, as visualized in Appendix B. Additionally, I will convert the `TractSNAP` continuous variable to a factor with 3 different levels of high, medium, and low number of households receiving SNAP benefits. As seen in the EDA interaction plot, the slopes for the poverty rate differ based on the levels of SNAP receipt, so binning them will ensure that the relationship is preserved rather than smoothed over when the variable is continuous.

To select an explanatory model, I calculated AIC and BIC for the following models.

1. Geographical Model: $$\log\left(\frac{\hat{\pi}_i}{1-\hat{\pi}_i}\right) = \hat{\beta}_0 + \hat{\beta}_1 \cdot \text{Urban}_i$$

2. Poverty Model: $$\begin{aligned}\log\left(\frac{\hat{\pi}_i}{1-\hat{\pi}_i}\right) = \hat{\beta}_0 &+ \hat{\beta}_1 \cdot \text{MedianFamilyIncome}_i \\ &+ \hat{\beta}_2 \cdot \text{bin_SNAP}_{\text{Mid},i} \\ &+ \hat{\beta}_3 \cdot \text{bin_SNAP}_{\text{High},i} \\ &+ \hat{\beta}_4 \cdot \text{HUNV}_i \\ &+ \hat{\beta}_5 \cdot \log(\text{PovertyRate}_i)\end{aligned}$$

3. Combined Model: $$\begin{aligned}\log\left(\frac{\hat{\pi}_i}{1-\hat{\pi}_i}\right) = \hat{\beta}_0 &+ \hat{\beta}_1 \cdot \text{Urban}_i \\ &+ \hat{\beta}_2 \cdot \text{MedianFamilyIncome}_i \\ &+ \hat{\beta}_3 \cdot \text{bin_SNAP}_{\text{Mid},i} \\ &+ \hat{\beta}_4 \cdot \text{bin_SNAP}_{\text{High},i} \\ &+ \hat{\beta}_5 \cdot \text{HUNV}_i \\ &+ \hat{\beta}_6 \cdot \log(\text{PovertyRate}_i)\end{aligned}$$

4. Interaction Model: $\log\left(\frac{\hat{\pi}_i}{1-\hat{\pi}_i}\right) = \hat{\beta}_0 + \hat{\beta}_1 \cdot \text{Urban}_i$

$$
\begin{aligned}
&+ \hat{\beta}_2 \cdot \text{MedianFamilyIncome}_i \\
&+ \hat{\beta}_3 \cdot \text{bin_SNAP}_{\text{Mid},i} \\
&+ \hat{\beta}_4 \cdot \text{bin_SNAP}_{\text{High},i} \\
&+ \hat{\beta}_5 \cdot \text{HUNV}_i \\
&+ \hat{\beta}_6 \cdot \log(\text{PovertyRate}_i) \\
&+ \hat{\beta}_7 \cdot (\text{bin_SNAP}_{\text{Mid},i} \times \log(\text{PovertyRate}_i)) \\
&+ \hat{\beta}_8 \cdot (\text{bin_SNAP}_{\text{High},i} \times \log(\text{PovertyRate}_i))
\end{aligned}
$$

where $\hat{\pi}_i$ represents the probability a census tract is a food desert

Testing each of these models allowed me to understand which interactions and predictors were most helpful in explaining the existence of a food desert census tract in NC. AIC/BIC revealed that the models with the lowest scores were the combined model (AIC: 254.14; BIC: 282.6027) and the interaction model (AIC: 252.6751; BIC: 289.27). Since the difference in AIC is too small to decide on, we can look at BIC. The difference in BIC between models is slightly larger at about 7, preferring the combined, more parsimonious model. However, I wish to refrain from relying on primarily BIC for model selection since it is better used to select predictive rather than explanatory models. Thus, I will also compare the AUC of the combined and interaction model. The AUC of the combined model is 0.8993 and the AUC of the interaction model is 0.9033. Since even AUC is very close, I look to using a drop in deviance test to understand whether the interaction term is a significant addition to the model. The test reveals a p-value of 0.065, which is close to the threshold of 0.05, and therefore, we would conclude that the interaction term does not have a significant effect on the outcome.

Though statisticians usually prefer to use the most parsimonious model, and the drop in devaince test and BIC reveal that the combined model is likely preferred, I ultimately choose the interaction model to be my final model. This is because I aim to prioritize explanatory power of the model and since the interaction effect has close to a signficant impact, the AIC is indistinguishable between the final two models, and the EDA shows a clear, interesting interplay between snap receipt and poverty rate, I decide that these indications are enough to justify the model with more predictors.

Evaluating the variance inflation factors for the predictors of the interaction model revealed substantial multicollinearity, as the interaction terms had VIF scores of about 70 and 92. Not to mention, even the individual levels of bin_SNAP had high VIF scores aroung 60 and 70. While some inflation is to be expected due to the interaction being correlated with the individual variables involved in the effect, these extremely high scores make it difficult to justify using

this model. To reduce multicollinearity, I will relevel bin_SNAP and use High SNAP as the baseline since it had a much different relationship with poverty rate than the other groups. Doing so revealed a substantial decrease in the VIF scores. Though they were still inflated around 30 to 39, these were not as conerning since some collinearity is expected when modeling interaction effects. Thus, the final model will include the releveled bin_SNAP.

Results

$$\begin{aligned}\text{Final Model: } \log\left(\frac{\hat{\pi}_i}{1-\hat{\pi}_i}\right) = \hat{\beta}_0 &+ \hat{\beta}_1 \cdot \text{Urban}_i \\ &+ \hat{\beta}_2 \cdot \text{MedianFamilyIncome}_i \\ &+ \hat{\beta}_3 \cdot \text{bin_SNAP}_{\text{Mid},i} \\ &+ \hat{\beta}_4 \cdot \text{bin_SNAP}_{\text{High},i} \\ &+ \hat{\beta}_5 \cdot \text{HUNV}_i \\ &+ \hat{\beta}_6 \cdot \log(\text{PovertyRate}_i) \\ &+ \hat{\beta}_7 \cdot (\text{bin_SNAP}_{\text{Mid},i} \times \log(\text{PovertyRate}_i)) \\ &+ \hat{\beta}_8 \cdot (\text{bin_SNAP}_{\text{High},i} \times \log(\text{PovertyRate}_i))\end{aligned}$$

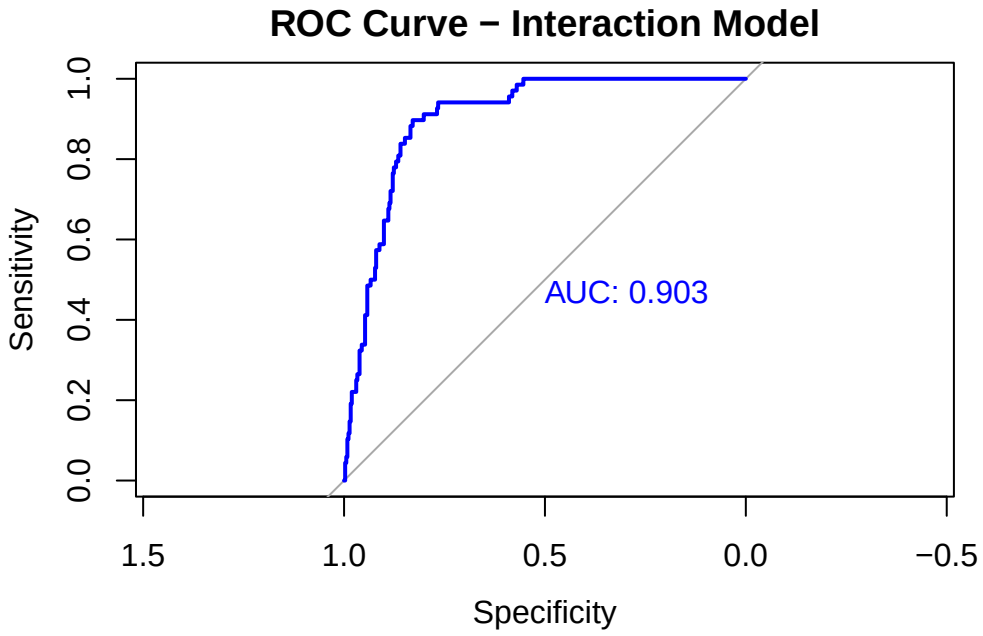
term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	13.01487	2.60802	0.98392	0.32515	0.08632	2476.12113
MedianFamilyIncome	0.99995	0.00002	-3.17000	0.00152	0.99991	0.99998
UrbanUrban	13.18194	0.46022	5.60350	0.00000	5.67035	35.13406
bin_SNAPLow SNAP	0.00067	2.89436	-2.52258	0.01165	0.00000	0.12079
bin_SNAPMid SNAP	0.00570	2.44548	-2.11287	0.03461	0.00004	0.58045
log(PovertyRate)	0.52997	0.68302	-0.92961	0.35257	0.13689	2.01674
TractHUNV	0.99732	0.00185	-1.45505	0.14565	0.99358	1.00085
bin_SNAPLow SNAP:log(PovertyRate)	8.22348	1.02286	2.05990	0.03941	1.28271	76.15941
bin_SNAPMid SNAP:log(PovertyRate)	3.30781	0.79596	1.50294	0.13285	0.72213	16.84570

Model assumptions and diagnostics

Logistic regression requires that observations are independent, that the log-odds of the outcome have a linear relationship with continuous predictors, and that there is no severe multicollinearity among predictors. The independence assumption is partially addressed by our random 20%

sample of North Carolina census tracts, which reduces (though does not eliminate) the spatial dependence among neighboring tracts. In Appendix B, I binned and plotted each quantitative predictor against the log odds of being a food desert and found a straight, linear relationship for each, except poverty rate. As a result, I log transformed poverty rate for each of the models I selected between. Regarding multicollinearity, while the VIF scores were inflated for the bin_SNAP and interaction terms (scores were around 30 to 39), this level of inflation is expected due to the interaction, and since the other predictors not involved in the interaction had scores less than 10, we can conclude that the model does not exhibit multicollinearity. Finally, Cook's distance confirmed no influential outliers that would affect model fit.

Model Fit



The ROC curve above demonstrates how well the final model distinguishes between the binary outcomes of being a food desert vs. not being a food desert. At an AUC of 0.903, we can conclude that the model distinguishes extremely well between classes. AUC is also robust to class imbalance, meaning it is not artificially inflated.

Interpretation

In the final model, `MedianFamilyIncome` , `Urban` , `bin_SNAPLow SNAP`, `bin_SNAPMid SNAP` and `bin_SNAPLow SNAP:log(PovertyRate)` are all found to be statistically significant indicators.

As expected, tracts with increased income levels are less likely to be food deserts. For every $1,000 increase in median family income, the odds of a tract being classified as a food desert decrease by 4.9%, and we are 95% confident that the decrease in odds lies between about 2% and 8.6%, holding all other variables constant. Additionally, the model shows that geographic factors, like whether a tract is rural or urban, have a significant impact on the classification of a tract as a food desert or not. An urban tract is 13.18 times more likely than a rural tract to be classified as a food desert, holding all other variables constant.

Finally, the signficance of the `bin_SNAPLow SNAP:log(PovertyRate)` term in the model (p value $0.03941 < 0.05$) provide confirmation that there is some significant relationship between these variables and supports my justification for ultimately including it in the model. The interaction term indicates that the relationship between poverty and food desert status differs by SNAP participation level. Specifically, for each one-unit increase in the log poverty rate, the odds that a census tract with low SNAP participation is classified as a food desert increase by a factor of 8.22 more than the corresponding change for a tract with high SNAP participation, holding all other variables constant. This suggests that rising poverty is associated with a substantially stronger increase in food desert risk in areas where fewer households receive SNAP benefits, largely confirming the patterns seen in the EDA interaction plot where. Taken independently, this term also shows that a 1 unit increase in log poverty rate in Low SNAP tracts is associated with about 4.36 times higher odds of being a food desert, holding other variables constant.

Overall, the model suggests that income, SNAP receipt, urban classification, and the relationship between SNAP receipt and poverty rate provide the most explanation of the factors influencing likelihood of a tract in NC being a food desert or not. Specifically, we see that higher income tracts are associated with a lower likelihood being food deserts, urban tracts are associated with a higher likelihood being a food desert than a rural tract. Finally, the significant interaction effect between poverty rate and SNAP participation indicates that increases in poverty are associated with a much stronger rise in food desert risk in tracts where fewer households receive SNAP benefits.

Discussion

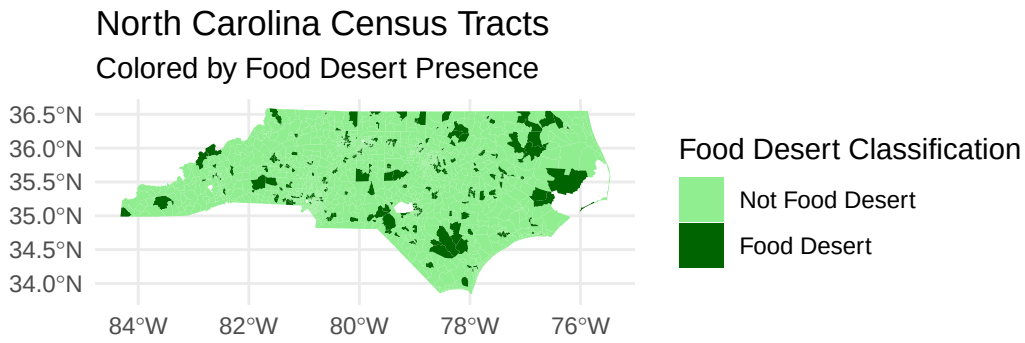
This analysis found that both economic and geographic factors like median family income, SNAP benefit receipt, and urbanicity are the strongest predictors of food desert classification among North Carolina census tracts. These results align with prior research that suggests neighborhood location and poverty rates are closely tied to food access limitations. It is especially important to note the finding that urban tracts in NC have 14.67 times the odds of being classified as food deserts compared to rural tracts because it challenges my hypothesis and the commonly held assumption that food deserts primarily occur in rural areas. This phenomenon points to urban disinvestment shaping food access. Furthermore, the relationship between SNAP participation and food desert status shows how government programs aimed at improving food access are insufficient at overcoming structural barriers to food access

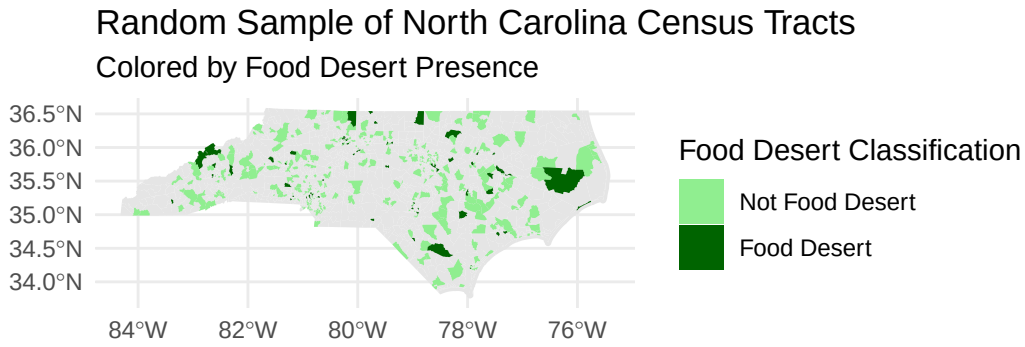
This study contains several limitations. Most notably, the 20% random sample reduces spatial correlation between tracts, but it is not a method to fully eliminate dependence among neighboring census tracts, which could affect the reliability of model predictions. Additionally, the class imbalance with only 16% of tracts classified as food deserts may limit the model's ability to detect important predictors of food desert status. Future work could address these limitations by using statistical methods to account for geographic clustering. Additionally, including variables relating to demographics like race, ethnicity, and age could illuminate the relevance of different categories of predictors — beyond economic or geographical. Finally, in attempting to create the most explanatory model, I act against the traditional practice of picking the most parsimonious model as a statistician because I strongly believed that it was important to capture the pattern seen in the EDA interaction plot despite negligible differences in AIC, nonsignificant results from the drop in deviance test, and a slightly influential indicator from BIC showing that the combined model would be better. Due to this modeling choices, it is likely I have inserted my own biased opinions into modeling, which could impact how well my model performs. However, we could look to ROC-AUC and show slight justification as it is marginally better at distinguishing between classes. Altogether, the difference between the two models is extremely small, if at all, and either model it seems would provide a great level of explanation about what factors contribute to the classification of a census tract as a food desert.

AI Disclosure

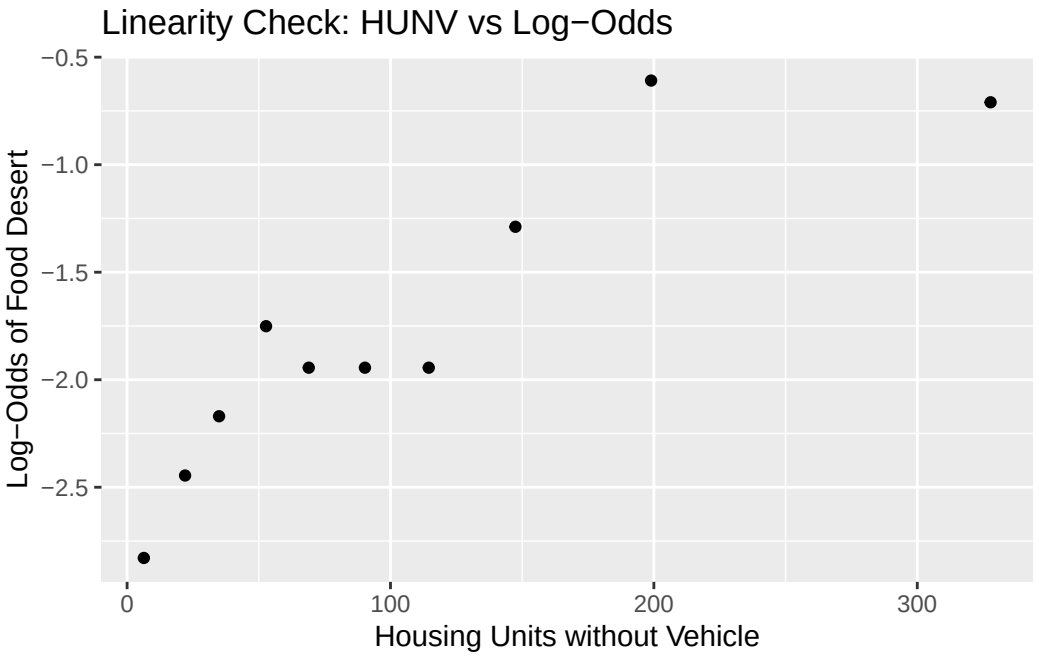
- I gave it my code and asked it to help me fix it.
- I asked it about an error or why code would do something I didn't want.
- Other: I asked it how I could make the linearity plots with the predictor binned and the output as log odds.

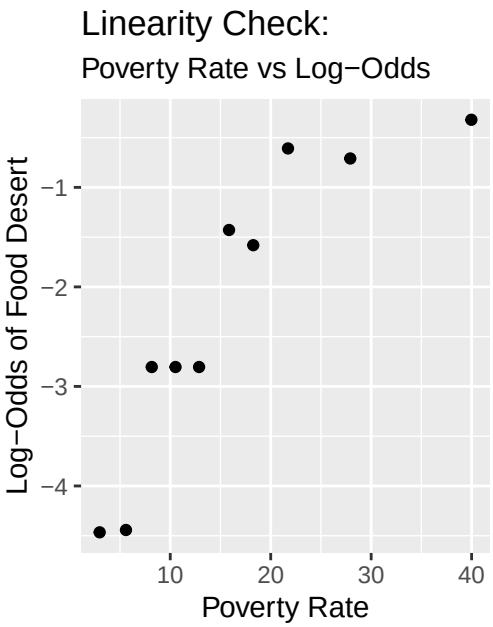
Appendix A

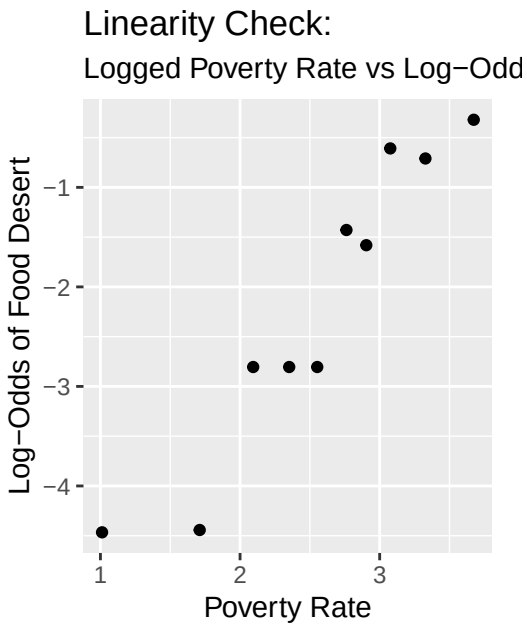

North Carolina Census Tracts
Colored by Food Desert Presence
36.5°N
36.0°N
35.5°N
35.0°N
34.5°N
34.0°N
84°W
82°W
80°W
78°W
76°W
Food Desert Classification
Not Food Desert
Food Desert

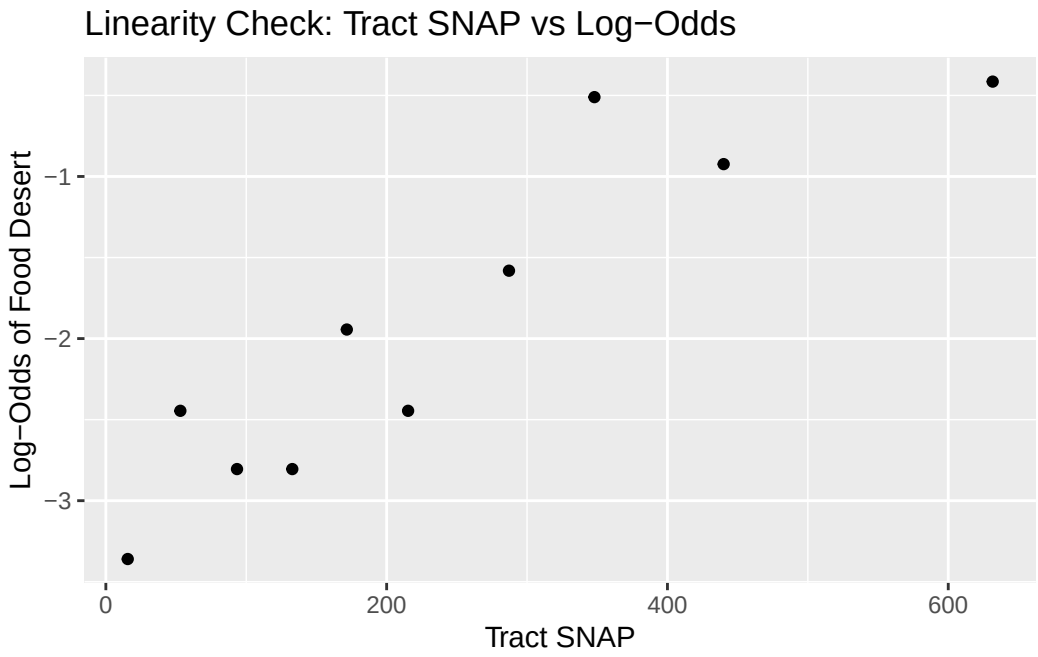

Random Sample of North Carolina Census Tracts
Colored by Food Desert Presence
36.5°N
36.0°N
35.5°N
35.0°N
34.5°N
34.0°N
84°W
82°W
80°W
78°W
76°W
Food Desert Classification
Not Food Desert
Food Desert

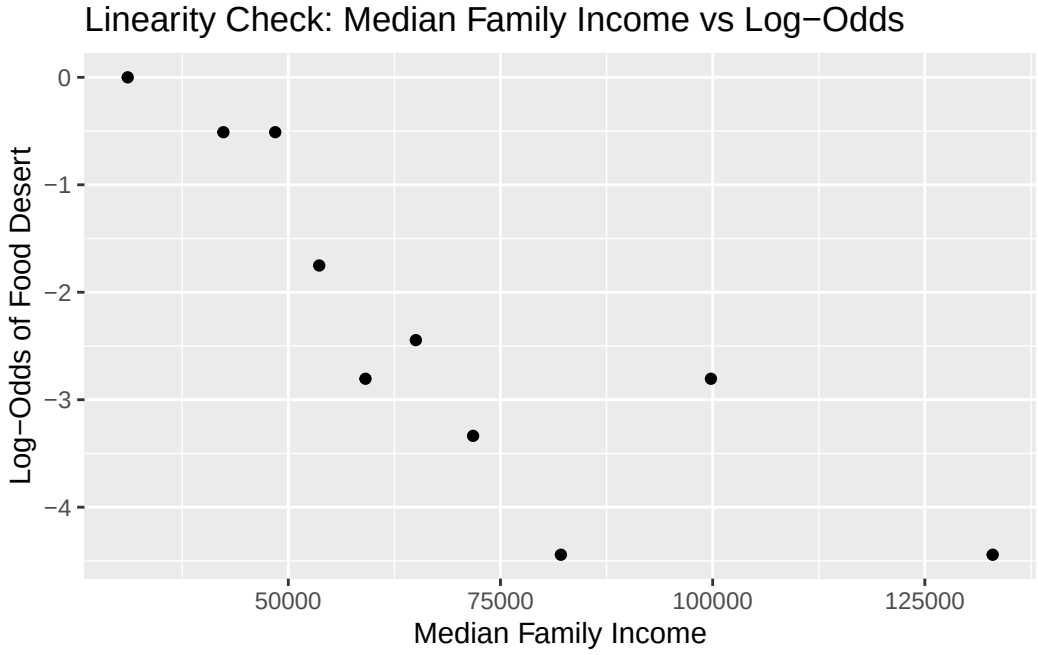
Appendix B


Linearity Check: HUNV vs Log−Odds
Log−Odds of Food Desert
−0.5
−1.0
−1.5
−2.0
−2.5
0
100
200
300
Housing Units without Vehicle


Linearity Check:
Poverty Rate vs Log−Odds
Log−Odds of Food Desert
−1
−2
−3
−4
10
20
30
40
Poverty Rate


Linearity Check:
Logged Poverty Rate vs Log−Odd
Log−Odds of Food Desert
−1
−2
−3
−4
1
2
3
Poverty Rate


Linearity Check: Tract SNAP vs Log−Odds
Log−Odds of Food Desert
−1
−2
−3
0
200
400
600
Tract SNAP


Linearity Check: Median Family Income vs Log−Odds
Log−Odds of Food Desert
0
−1
−2
−3
−4
50000
75000
100000
125000
Median Family Income